AUC And ROC curve

In classification, there are many different evaluation metrics. The most popular is accuracy, which measures how often the model is correct. This is a great metric because it is easy to understand and getting the most correct guesses is often desired. There are some cases where you might consider using another evaluation metric.

Another common metric is AUC, area under the receiver operating characteristic (ROC) curve. The Reciever operating characteristic curve plots the true positive (TP) rate versus the false positive (FP) rate at different classification thresholds.

The thresholds are different probability cutoffs that separate the two classes in binary classification. It uses probability to tell us how well a model separates the classes.

Imbalanced Data

Suppose we have an imbalanced data set where the majority of our data is of one value. We can obtain high accuracy for the model by predicting the majority class.

Although we obtain a very high accuracy, the model provided no information about the data so it's not useful. We accurately predict class 1 100% of the time while inaccurately predict class 0 0% of the time. At the expense of accuracy, it might be better to have a model that can somewhat separate the two classes.

In cases like this, using another evaluation metric like AUC would be preferred.

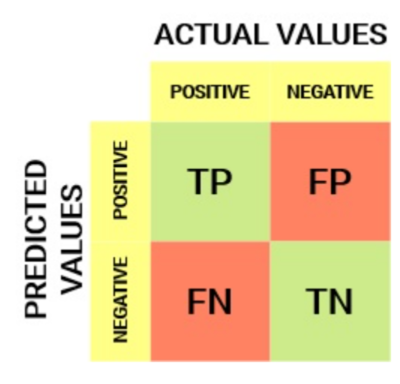
ROC and AUC demistyfied

You can use ROC (Receiver Operating Characteristic) curves to evaluate different thresholds for classification machine learning problems.

In a nutshell, ROC curve visualizes a confusion matrix for every threshold.

Now if talk about confusion matrix , we should be aware about some terminologies.

What are Sensitivity and Specificity?



Sensitivity / True Positive Rate / Recall

Sensitivity formula

$$Sensitivity = \frac{TP}{TP + FN}$$

Sensitivity tells us what proportion of the positive class got correctly classified.

A simple example would be determining what proportion of the actual sick people were correctly detected by the model.

False Negative Rate

False Negative Rate

$$FNR = \frac{FN}{TP + FN}$$

False Negative Rate (FNR) tells us what proportion of the positive class got incorrectly classified by the classifier.

A higher TPR and a lower FNR are desirable since we want to classify the positive class correctly.

Specificity / True Negative Rate

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$$Specificity = \frac{TN}{TN + FP}$$

Specificity tells us what proportion of the negative class got correctly classified.

Taking the same example as in Sensitivity, Specificity would mean determining the proportion of healthy people who were correctly identified by the model.

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$$FPR = \frac{FP}{TN + FP} = 1 - Specificity$$

FPR tells us what proportion of the negative class got incorrectly classified by the classifier.

A higher TNR and a lower FPR are desirable since we want to classify the negative class correctly.

Out of these metrics, Sensitivity and Specificity are perhaps the most important, and we will see later on how these are used to build an evaluation metric. But before that, let's understand why the probability of prediction is better than predicting the target class directly.

Probability of Predictions

A machine learning classification model can be used to directly predict the data point's actual class or predict its probability of belonging to different classes. The latter(probability part) gives us more control over the result. We can determine our own threshold to interpret the result of the classifier. This is sometimes more prudent than just building a completely new model!

Setting different thresholds for classifying positive classes for data points will inadvertently change the Sensitivity and Specificity of the model. And one of these thresholds will probably give a better result than the others, depending on whether we are aiming to lower the number of False Negatives or False Positives.

Have a look at the table below:

ID	Actual	Prediction Probability	>0.6	>0.7	> 0.8	Metric
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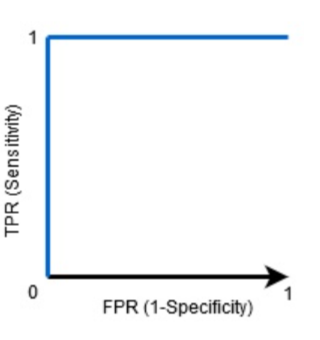
9	1	0.82	1	1	1	
10	0	0.86	1	1	1	
			0.75	0.5	0.5	TPR
			1	1	0.66	FPR
			0	0	0.33	TNR
			0.25	0.5	0.5	FNR

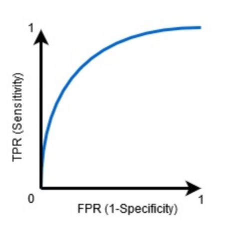
The metrics change with the changing threshold values. We can generate different confusion matrices and compare the various metrics that we discussed in the previous section. But that would not be a prudent thing to do. Instead, we can plot roc curves between some of these metrics to quickly visualize which threshold is giving us a better result. The AUC-ROC curve solves just that problem!

The Receiver Operator Characteristic (ROC) curve is an evaluation metric for binary classification problems. It is a probability curve that plots the TPR against FPR at various threshold values and essentially separates the 'signal' from the 'noise.' In other words, it shows the performance of a classification model at all classification thresholds. The Area Under the Curve (AUC) is the measure of the ability of a binary classifier to distinguish between classes and is used as a summary of the ROC curve.

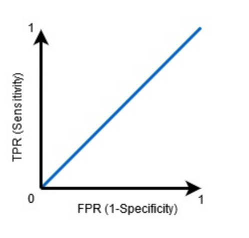
The higher the AUC, the better the model's performance at distinguishing between the positive and negative classes.



When AUC = 1, the classifier can correctly distinguish between all the Positive and the Negative class points. If, however, the AUC had been 0, then the classifier would predict all Negatives as Positives and all Positives as Negatives.

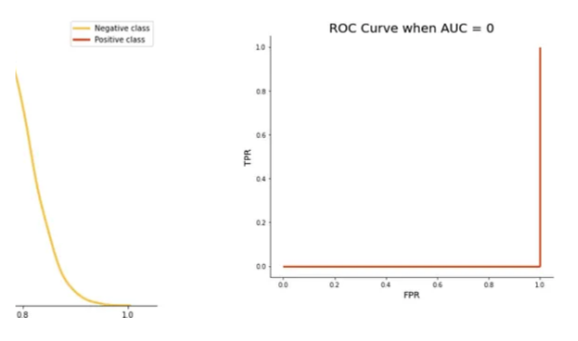


When 0.5<AUC<1, there is a high chance that the classifier will be able to distinguish the positive class values from the negative ones. This is so because the classifier is able to detect more numbers of True positives and True negatives than False negatives and False positives.



When AUC=0.5, then the classifier is not able to distinguish between Positive and Negative class points. Meaning that the classifier either predicts a random class or a constant class for all the data points. Just think about it, you ask a model whether someone is positive or negative, and it tells you: well, maybe it's positive, maybe it's negative (50:50). That's useless for binary classification tasks.

Can AUC be 0? Yes — it means the model is reciprocating the classes. In other words, it's predicting positive classes and negative and vice versa. Take a look at the image below:

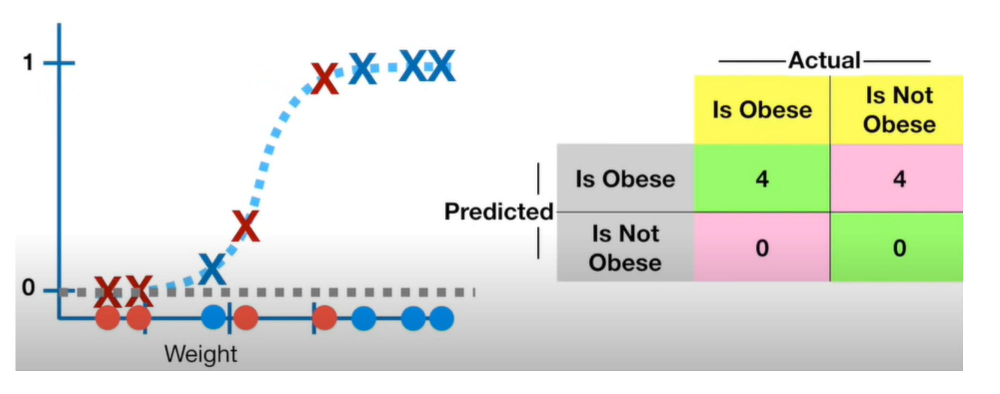


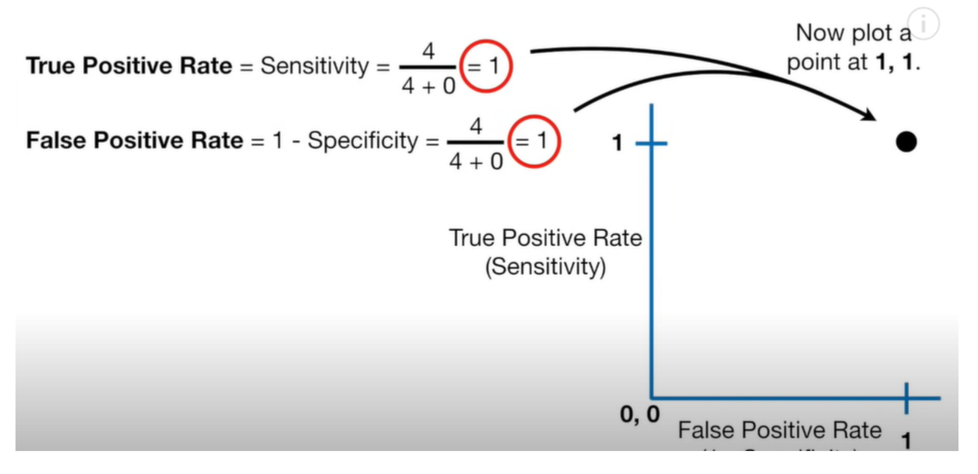
So, the higher the AUC value for a classifier, the better its ability to distinguish between positive and negative classes.

How Does the AUC-ROC Curve Work?

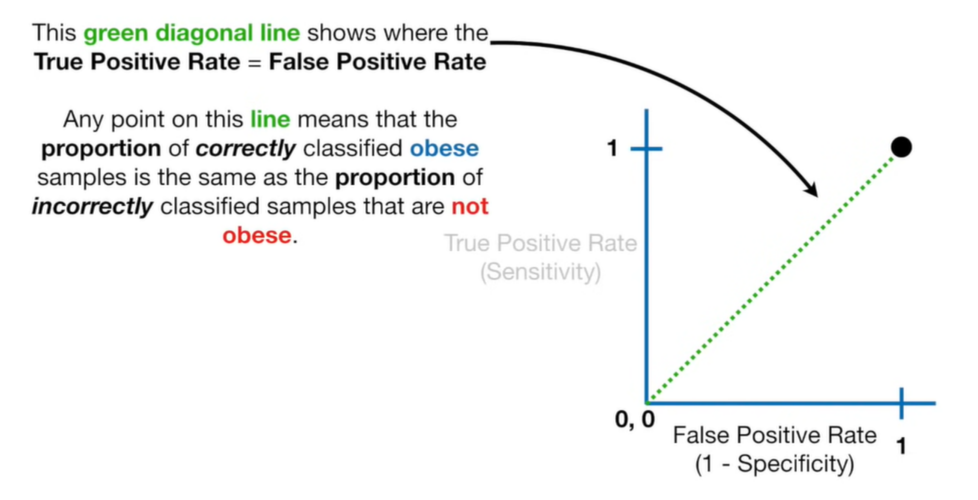
In a ROC curve, a higher X-axis value indicates a higher number of False positives than True negatives. While a higher Y-axis value indicates a higher number of True positives than False negatives. So, the choice of the threshold depends on the ability to balance False positives and False negatives.

Suppose we get this confusion matrix on taking threshold as 0.1

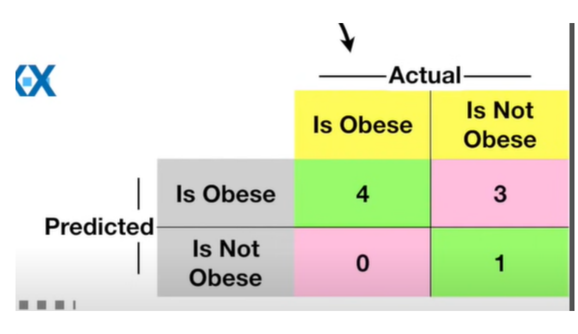


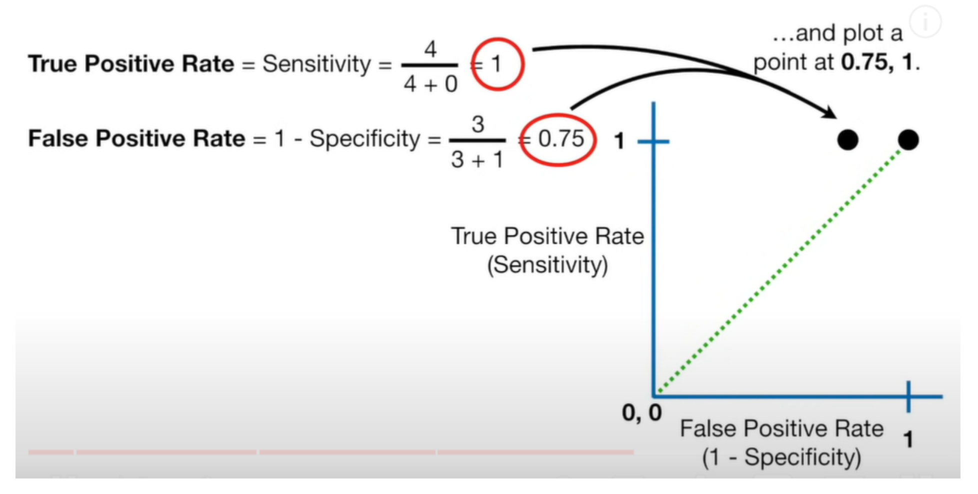


A point at **1, 1** means that even though we correctly classified ***all*** of the **obese** samples, we *incorrectly* classified ***all*** of the samples that were **not obese**.



Now let us make a confusion matrix with other threshold value

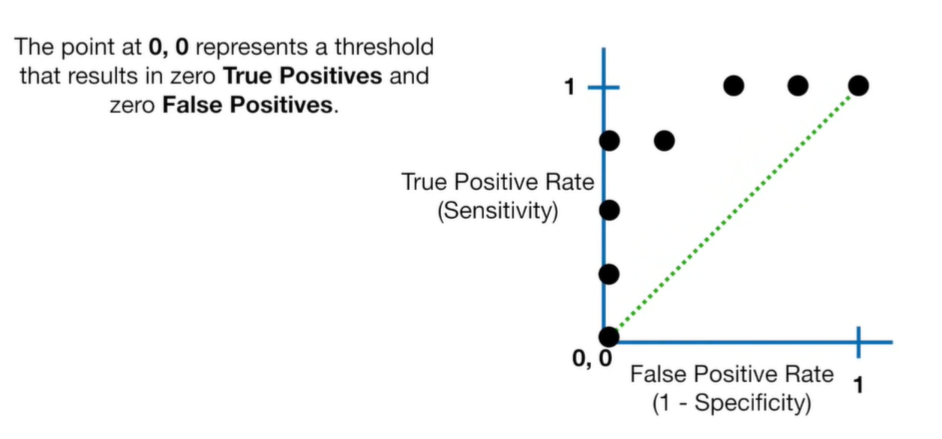


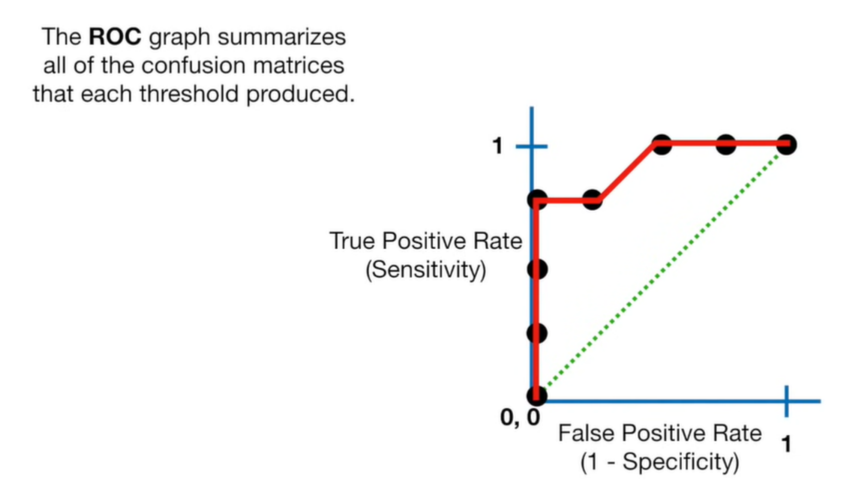


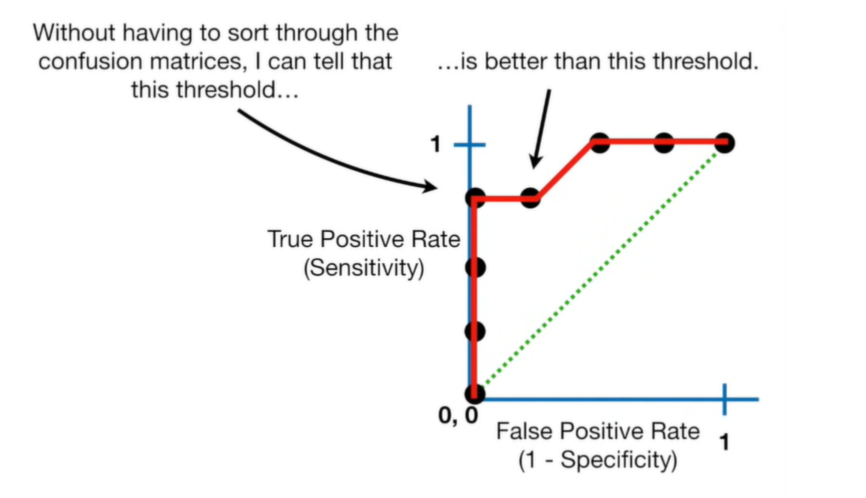
Since the new point (**0.75, 1**) is to the left of the **dotted green line**, we know that the proportion of correctly classified samples that were **obese** (true positives) *is greater* than the proportion of the samples that were incorrectly classified as **obese** (false positives).

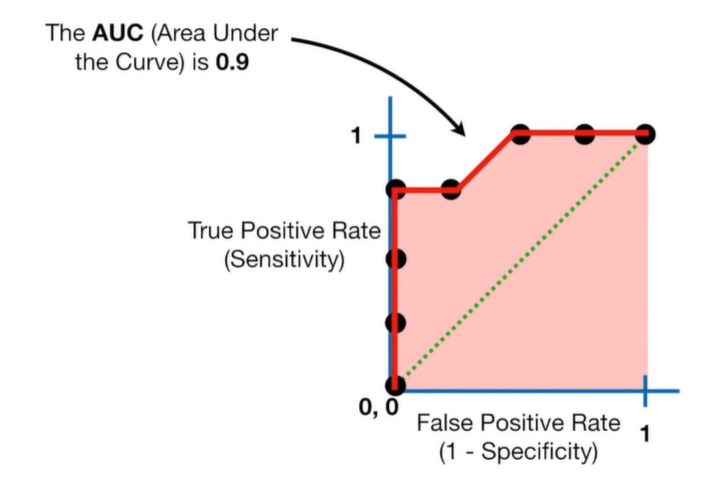
In other words, the new threshold for deciding if a sample is **obese** or **not** is better than the first one.

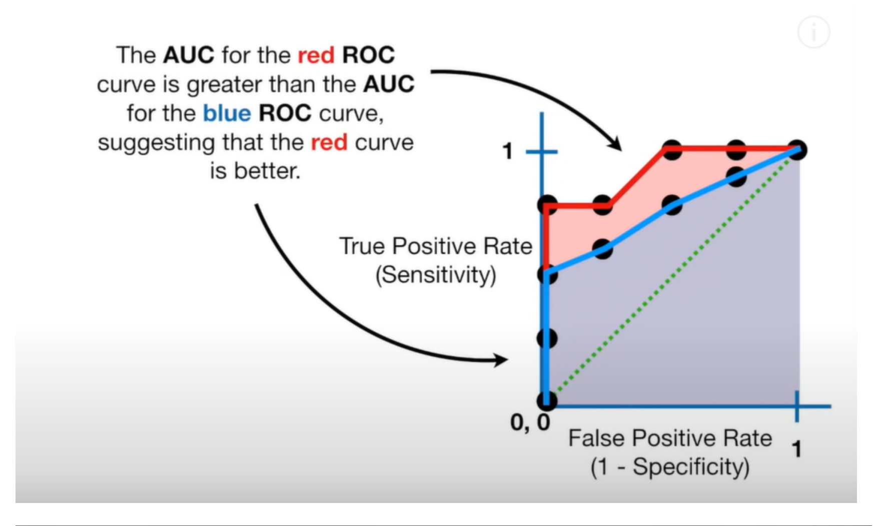
similarly we all other threshold values we got graph like this

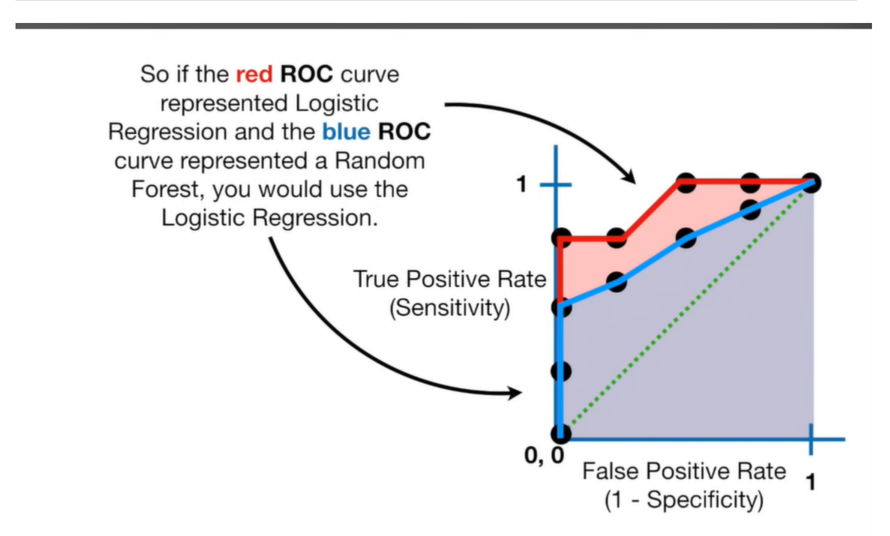












Summary : An AUC score of around .5 would mean that the model is unable to make a distinction between the two classes and the curve would look like a line with a slope of 1. An AUC score closer to 1 means that the model has the ability to separate the two classes and the curve would come closer to the top left corner of the graph.

AUC-ROC Curve for Multi-Class Classification

As I said before, the AUC-ROC curve is only for binary classification problems. But we can extend it to multiclass classification problems using the One vs. All technique.

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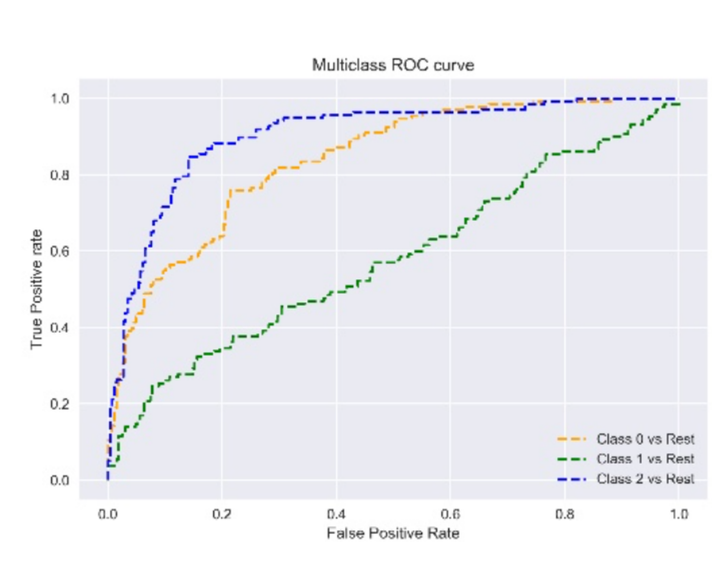

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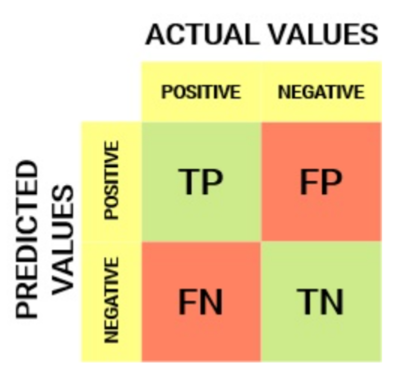

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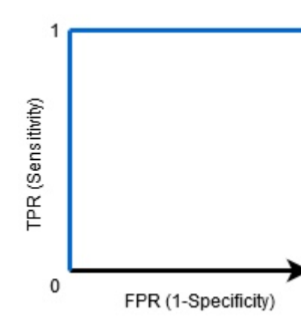

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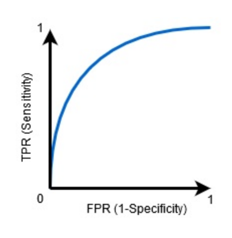

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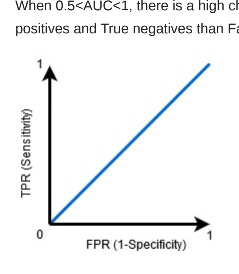

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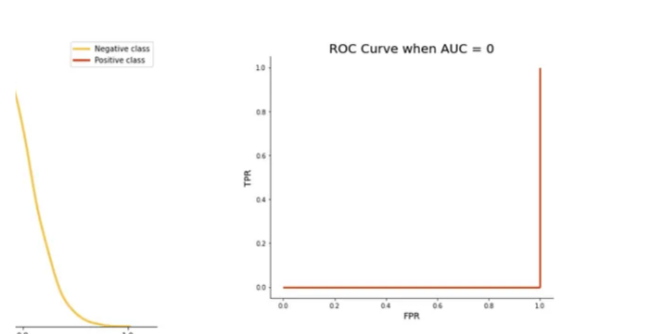

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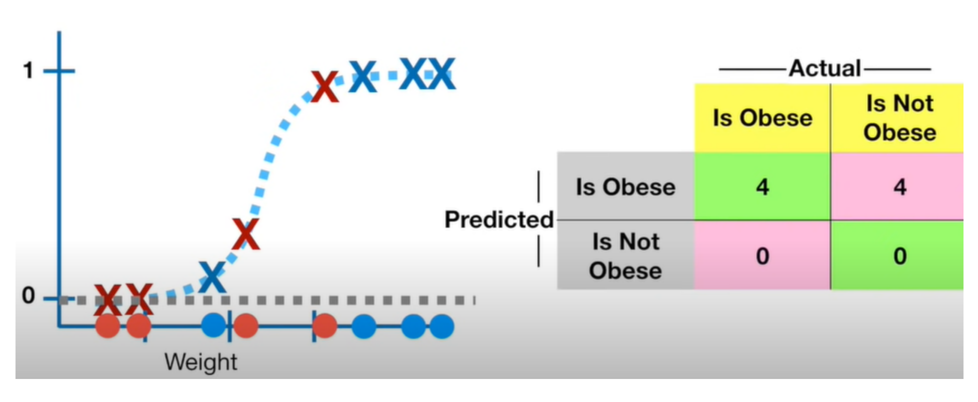

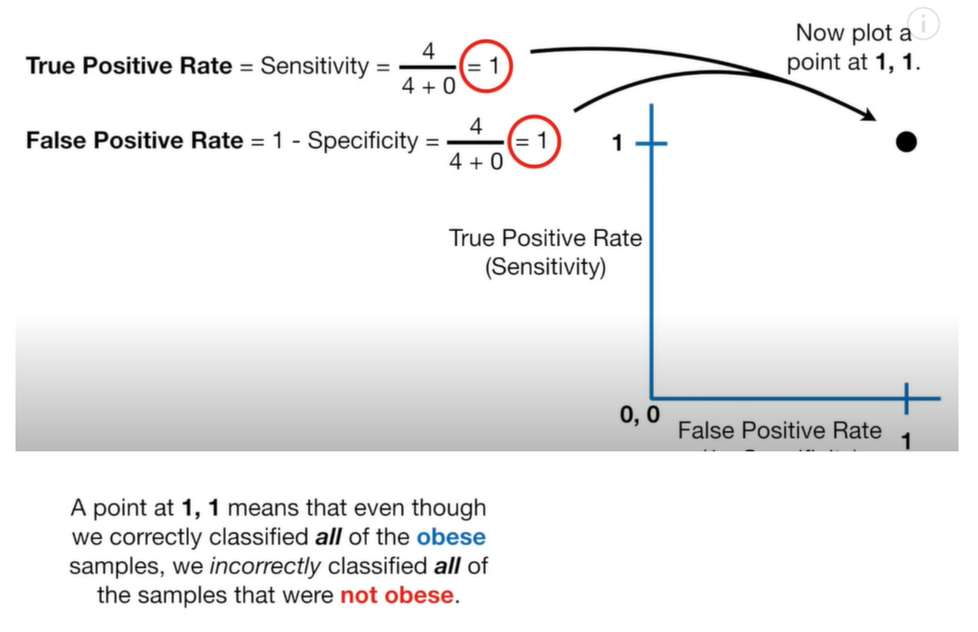

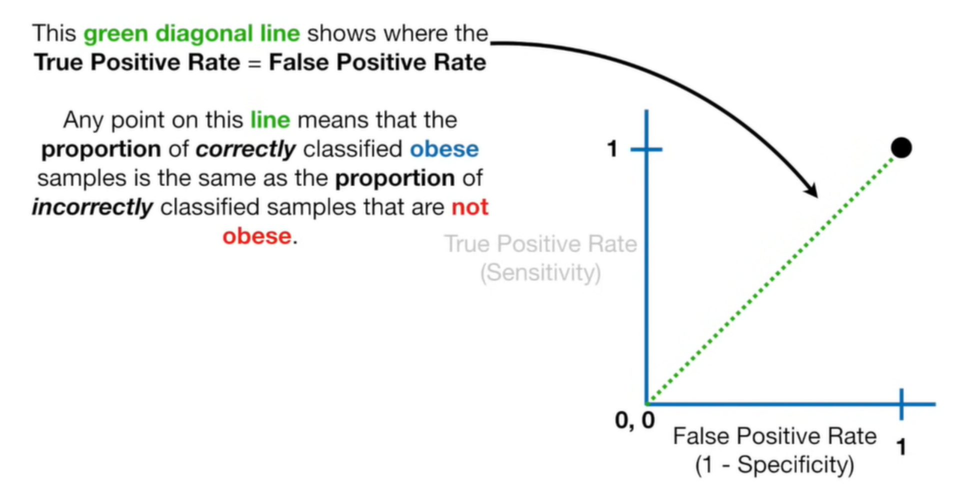

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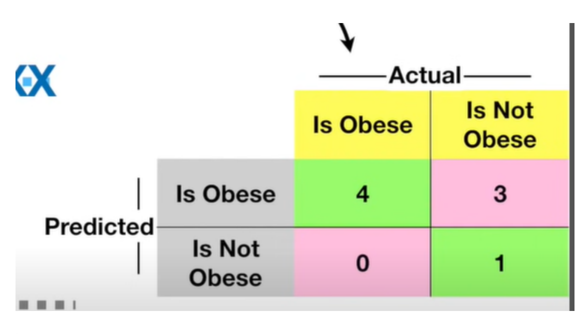

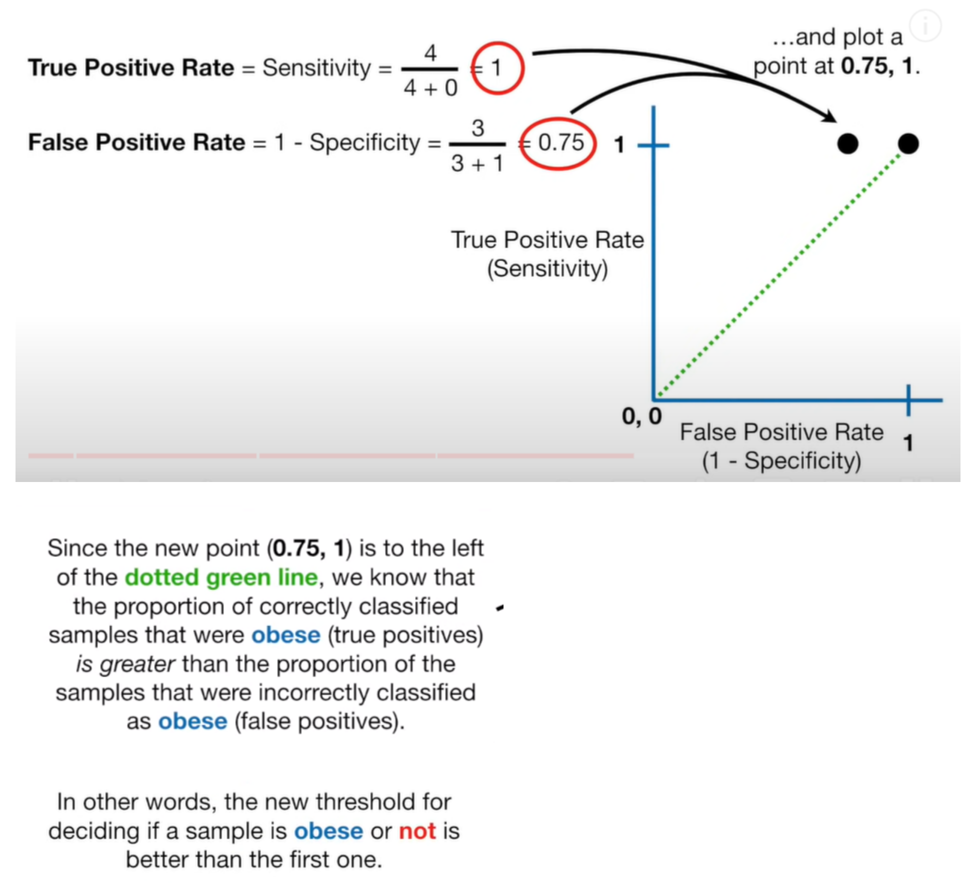

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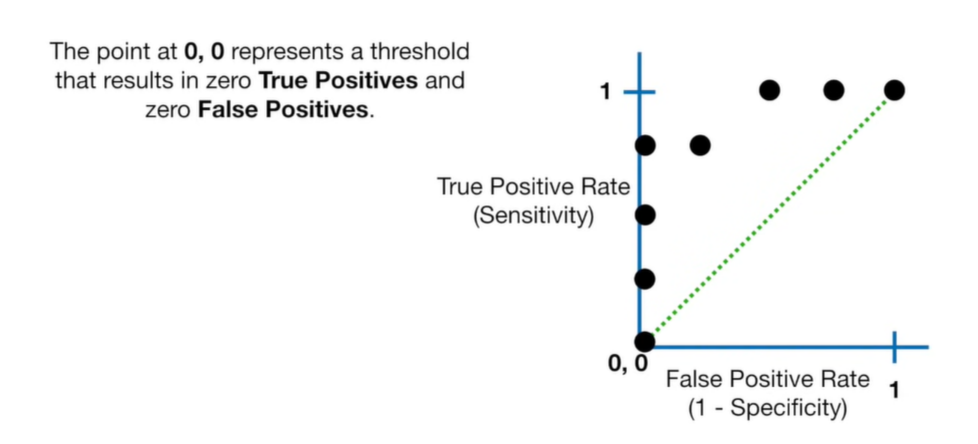

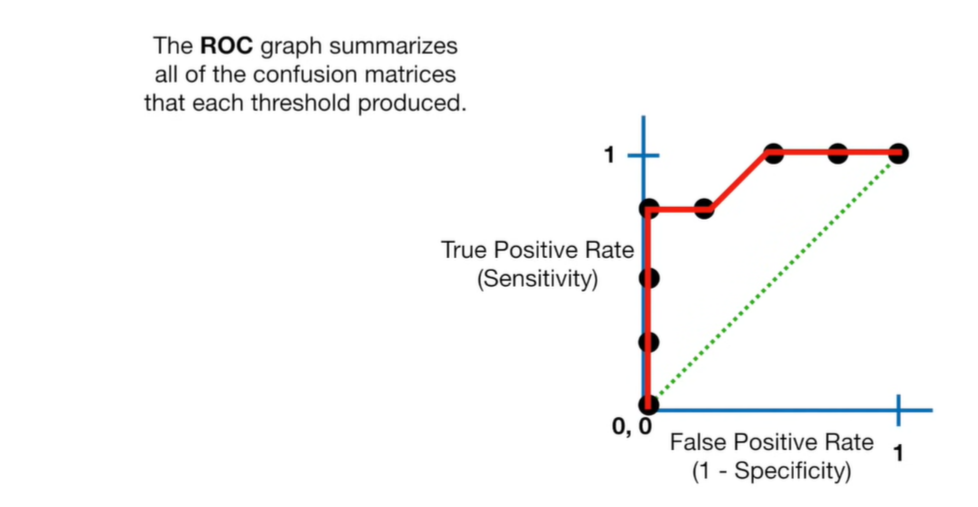

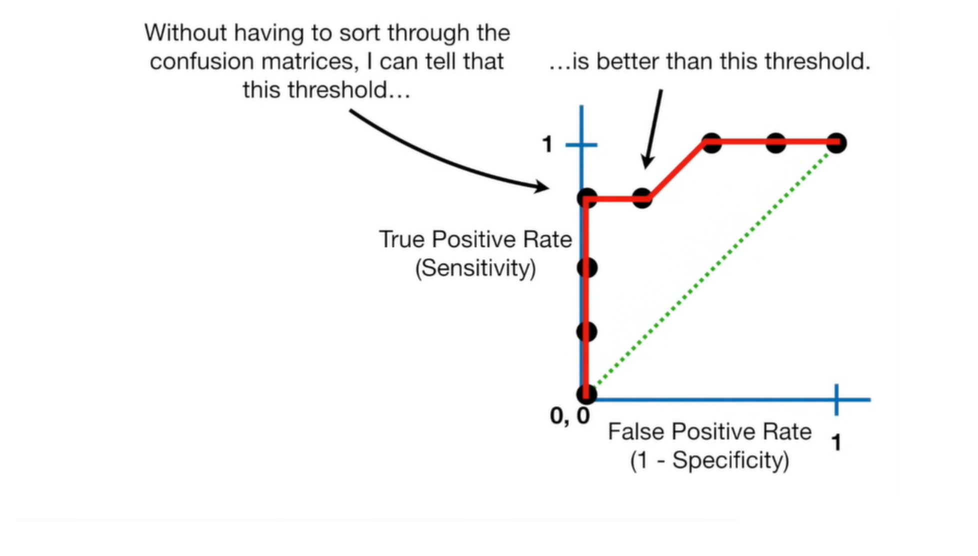

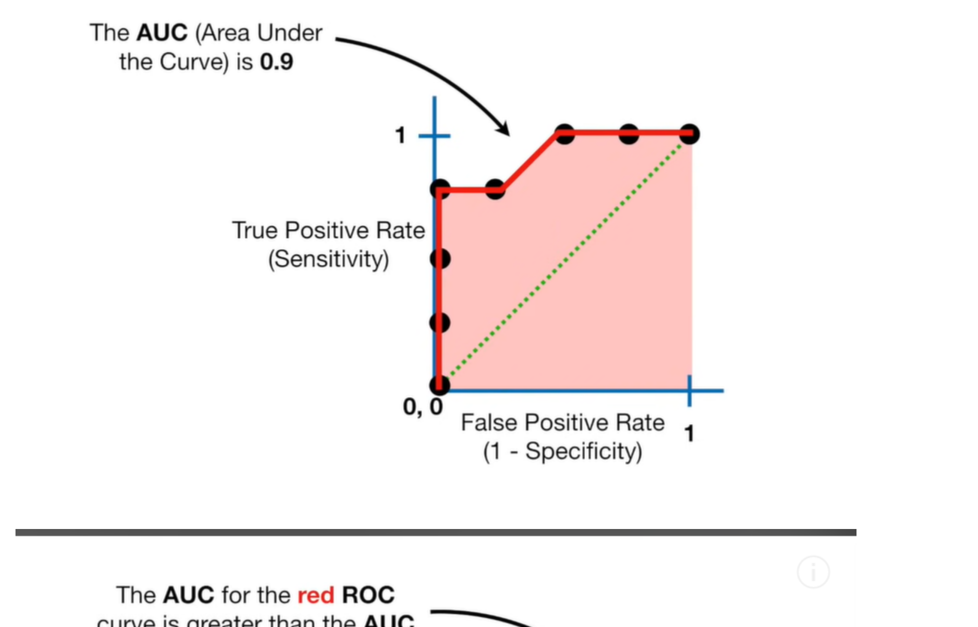

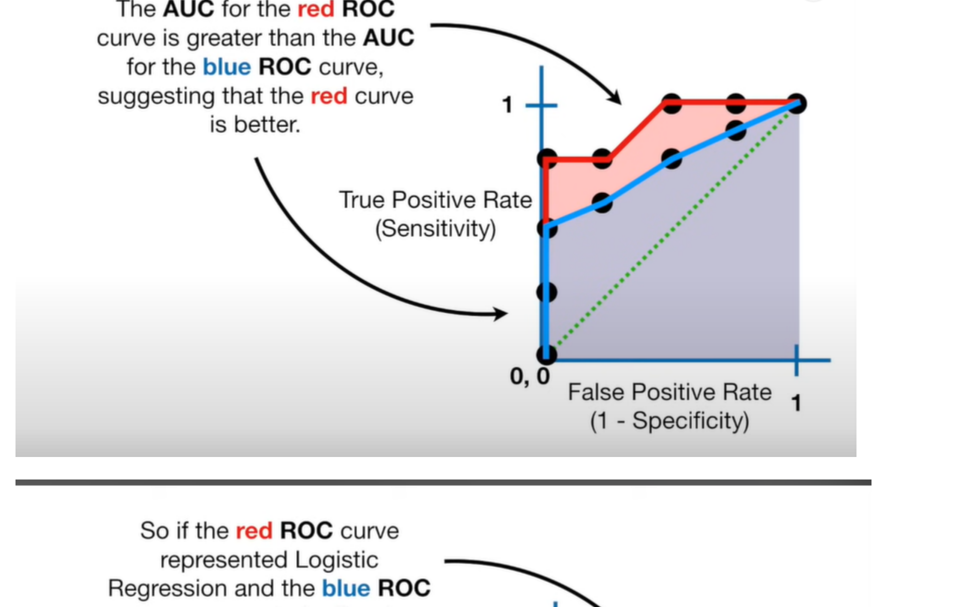

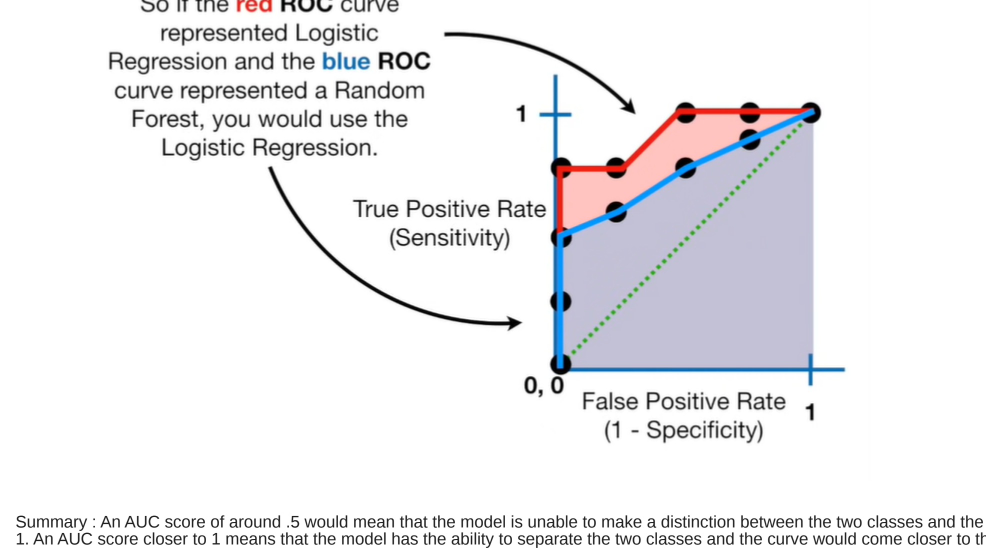

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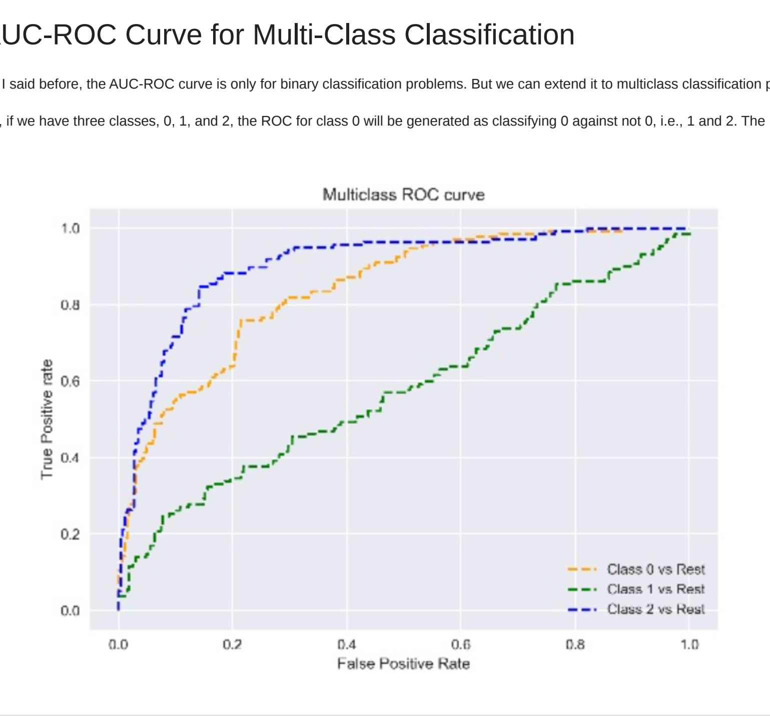

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